Ex.-No:8: Develop a Program to create a Reverse Shell using TCP SOCKETS.

13/08/26

AIM:

The aim of this Exercise is to understand and implement shell using TCP sockets. This will demonstrate how a client can connect back to an attacker (server) machine and allow the attacker to remotely execute commands on the client machine

DESCRIPTION:

A reverse shell is a type of shell in which the target machine connects back to the attacking machine and allows the attacker to execute commands remotely. In this Exercise we will simulate a basic reverse shell using Python's TCP socket library.

TOOLS & TECHNOLOGIES:

- Programming Language: Python 3x
- Libraries
- Socket for network communication via TCP/IP Sockets.

LAB SETUP

1. Server Machine (Attacker)
   - IP Address: The Machine that will listen for incoming connections from the client

PROCEDURE:

Step 1: Set up the Server (listener) Script on the Attacker ~~on~~ machine!

Create a new Python script on the attacker machine (eg: Server.py)

2. Copy the following code into the script:

```
import socket import subprocess import
os
Server_IP = 0.0.0.0
Server_Port = 9999
Socket.SOCK_STREAM)
```

3. Run the Server Script: Open a terminal an Run:

```
Python3 server.py
```

- The server will now listen for (connections on 0.0.0.0 (all interfaces) and Port 9999

4. Observe the output on the attackers terminal (server side). Each command will be Executed remotely on the client machine. and the results will be displayed on the Server.

OUTPUTS :

1. Server Side Output

. The server successfully listens on the specified port and waits for an incoming connection from the client.

Shell > pwd /home/user

Shell > cd /home

Shell > ls user

another_user

Shell > exit

[*] Closing connection

Result :

The client successfully connected to the server and executed commands sent by the attacker, returning their output.